DEEP LEARNING LABORATORY

Experiment 1 - Three-Layer Neural Network from Scratch

A Laboratory Experiment Report

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Dataset
MNIST

Prepared from the captured experiment notebook outputs

Experiment Overview

Experiment: Experiment 01

Dataset: MNIST

Three-Layer Neural Network from Scratch

This is a standalone notebook for this lab section. Run the setup and dataset cells first, then run the experiment cell and its output check. Use a Colab GPU runtime for the image and generative-model experiments when available.

Step 1 - Install the Additional Colab Packages

Wait until installation completes before the next cell starts.

Code

```
%pip -q install -U datasets "transformers<5" sentencepiece accelerate torchmetrics torch-fidelity ucimlrepo tabulate
```

Step 2 - Verify the Runtime

This cell records the Python, TensorFlow, and PyTorch versions and enables safe TensorFlow GPU memory growth.

Code

```
from pathlib import Path
import platform
import tensorflow as tf
import torch

project_root = Path("/content/Deeplearning")
project_root.mkdir(parents=True, exist_ok=True)
for gpu in tf.config.list_physical_devices("GPU"):
    try:
        tf.config.experimental.set_memory_growth(gpu, True)
    except RuntimeError:
        pass
print(f"Python: {platform.python_version()}")
print(f"TensorFlow: {tf.__version__}")
print(f"PyTorch: {torch.__version__}")
print(f"CUDA available: {torch.cuda.is_available()}")
print(f"Project root: {project_root}")
```

```
Python: 3.13.15
TensorFlow: 2.20.0
PyTorch: 2.11.0+cu128
CUDA available: True
Project root: /content/Deeplearning
```

Step 3 - Import Every Required Dataset

This cell creates visible files under `/content/Deeplearning/datasets` and prints a manifest that can be shown to the staff.

- MNIST, Iris, CIFAR-10, Dogs vs. Cats, Airline Passengers, and Tiny Shakespeare are downloaded as complete or balanced lab-ready files.

- MS COCO, CelebA, and LSUN Bedrooms are the genuine named datasets, streamed as practical subsets for Colab.

- No synthetic replacement dataset is used.

Code

```
from pathlib import Path
import json
import shutil
import numpy as np
```

```
import pandas as pd
import tensorflow as tf
from PIL import Image
from datasets import load_dataset
from sklearn.datasets import load_iris


root = Path("/content/Deeplearning") if Path("/content").exists() else Path(__file__).resolve().parent
dataset_dir = root / "datasets"
output_dir = root / "outputs"
dataset_dir.mkdir(parents=True, exist_ok=True)
output_dir.mkdir(parents=True, exist_ok=True)


def balanced_indices(labels, per_class, seed):
    labels = np.asarray(labels).reshape(-1)
    indices = np.concatenate([np.flatnonzero(labels == class_index)[:per_class] for class_index in np.unique(labels)])
    rng = np.random.default_rng(seed)
    rng.shuffle(indices)
    return indices


def save_image_stream(dataset_name, split, destination, count, size, config=None):
    destination.mkdir(parents=True, exist_ok=True)
    existing = sorted(destination.glob("*.jpg"))
    if len(existing) >= count:
        return len(existing)
    arguments = {
        "path": dataset_name,
        "split": split,
        "streaming": True,
    }
    if config is not None:
        arguments["name"] = config
    stream = load_dataset(**arguments)
    saved = 0
    for row in stream.take(count):
        image = row["image"].convert("RGB")
        image = image.resize((size, size), Image.Resampling.LANCZOS)
        image.save(destination / f"image_{saved:04d}.jpg", quality=92)
        saved += 1
    if saved < count:
        raise RuntimeError(f"Only {saved} images were available from {dataset_name}")
    return saved


(mnist_x_train, mnist_y_train), (mnist_x_test, mnist_y_test) = tf.keras.datasets.mnist.load_data()
mnist_train_indices = balanced_indices(mnist_y_train, 1000, 42)
mnist_test_indices = balanced_indices(mnist_y_test, 200, 42)
np.savez_compressed(
    dataset_dir / "mnist_lab.npz",
    x_train=mnist_x_train[mnist_train_indices],
    y_train=mnist_y_train[mnist_train_indices],
    x_test=mnist_x_test[mnist_test_indices],
    y_test=mnist_y_test[mnist_test_indices],
)

try:
    from ucimlrepo import fetch_ucirepo

    iris_repository = fetch_ucirepo(id=53)
    iris_features = iris_repository.data.features.copy()
    iris_features.columns = ["sepal_length", "sepal_width", "petal_length", "petal_width"]
    iris_target = iris_repository.data.targets.iloc[:, 0].astype(str)
    iris_target = iris_target.str.replace("Iris-", "", regex=False).str.replace("Iris ", "", regex=False).str.lower()
    iris = iris_features.copy()
    iris["species"] = iris_target.to_numpy()
    iris_source = "UCI Machine Learning Repository dataset 53"
except Exception:
    iris_bundle = load_iris(as_frame=True)
    iris = iris_bundle.data.copy()
    iris.columns = ["sepal_length", "sepal_width", "petal_length", "petal_width"]
    iris["species"] = pd.Series(iris_bundle.target).map(dict(enumerate(iris_bundle.target_names)))
    iris_source = "scikit-learn packaged Iris fallback"
iris.to_csv(dataset_dir / "iris.csv", index=False)

(cifar_x_train, cifar_y_train), (cifar_x_test, cifar_y_test) = tf.keras.datasets.cifar10.load_data()
cifar_train_indices = balanced_indices(cifar_y_train, 800, 42)
cifar_test_indices = balanced_indices(cifar_y_test, 200, 42)
np.savez_compressed(
    dataset_dir / "cifar10_lab.npz",
    x_train=cifar_x_train[cifar_train_indices],
    y_train=cifar_y_train[cifar_train_indices].reshape(-1),
    x_test=cifar_x_test[cifar_test_indices],
    y_test=cifar_y_test[cifar_test_indices].reshape(-1),
)
```

```
def build_cats_and_dogs_filtered(destination, image_size=160, seed=42):
    """Rebuild the classic 'cats_and_dogs_filtered' train/validation layout.

    The original tutorial archive at
    https://storage.googleapis.com/mledu-datasets/cats_and_dogs_filtered.zip
    now returns HTTP 403 (Google retired the mledu-datasets bucket), so the
    same 3,000-image split is reconstructed by streaming labelled images
    from the public "microsoft/cats_vs_dogs" dataset on the Hugging Face Hub.
    """
    targets = {
        destination / "train" / "cats": (0, 1000),
        destination / "train" / "dogs": (1, 1000),
        destination / "validation" / "cats": (0, 500),
        destination / "validation" / "dogs": (1, 500),
    }
    already_done = all(
        len(list(folder.glob("*.jpg"))) >= count for folder, (_, count) in targets.items()
    )
    if already_done:
        return destination
    for folder in targets:
        folder.mkdir(parents=True, exist_ok=True)
    counts = {folder: 0 for folder in targets}
    stream = load_dataset("microsoft/cats_vs_dogs", split="train", streaming=True)
    stream = stream.shuffle(seed=seed, buffer_size=2000)
    for row in stream:
        if all(counts[folder] >= count for folder, (_, count) in targets.items()):
            break
        label = row["labels"]
        for folder, (target_label, target_count) in targets.items():
            if label != target_label or counts[folder] >= target_count:
                continue
            try:
                image = row["image"].convert("RGB")
            except Exception:
                break
            image = image.resize((image_size, image_size), Image.Resampling.LANCZOS)
            image.save(folder / f"image_{counts[folder]:04d}.jpg", quality=92)
            counts[folder] += 1
            break
    for folder, (_, count) in targets.items():
        if counts[folder] < count:
            raise RuntimeError(f"Only {counts[folder]} images were available for {folder}")
    return destination


dogs_cats_dir = dataset_dir / "cats_and_dogs_filtered"
build_cats_and_dogs_filtered(dogs_cats_dir)
if not (dogs_cats_dir / "train").exists():
    raise FileNotFoundError("Dogs vs. Cats extraction folder was not created")

airline_url = "https://raw.githubusercontent.com/jbrownlee/Datasets/refs/heads/master/monthly-airline-passengers.csv"
airline = pd.read_csv(airline_url)
airline.to_csv(dataset_dir / "monthly_airline_passengers.csv", index=False)

shakespeare_cache = tf.keras.utils.get_file(
    fname="shakespeare.txt",
    origin="https://storage.googleapis.com/download.tensorflow.org/data/shakespeare.txt",
    cache_dir=str(root),
    cache_subdir="downloads",
)
shakespeare_path = dataset_dir / "shakespeare.txt"
shutil.copyfile(shakespeare_cache, shakespeare_path)

coco_dir = dataset_dir / "ms_coco_subset"
coco_dir.mkdir(parents=True, exist_ok=True)
coco_caption_path = coco_dir / "captions.csv"
if not coco_caption_path.exists() or len(list(coco_dir.glob("coco_*.jpg"))) < 5:
    coco_stream = load_dataset("mm-eval/MS-COCO-Captions", split="val", streaming=True)
    coco_rows = []
    for index, row in enumerate(coco_stream.take(5)):
        image_path = coco_dir / f"coco_{index:03d}.jpg"
        media = row["media"]
        image = media[0] if isinstance(media, list) else media
        image.convert("RGB").save(image_path, quality=92)
        coco_rows.append(
            {
                "file": image_path.name,
                "captions_json": json.dumps(list(row["answer"]), ensure_ascii=False),
            }
        )
    pd.DataFrame(coco_rows).to_csv(coco_caption_path, index=False)

celeba_dir = dataset_dir / "celeba_subset"
celeba_count = save_image_stream("flwrlabs/celeba", "train", celeba_dir, 384, 64, "img_align+identity+attr")
```

```
lsun_dir = dataset_dir / "lsun_bedrooms_subset"
lsun_count = save_image_stream("pcuenq/lsun-bedrooms", "train", lsun_dir, 512, 64)

manifest = pd.DataFrame(
    [
        {"dataset": "MNIST", "path": "mnist_lab.npz", "records": 12000, "experiments": "1", "source": "TensorFlow Keras
            MNIST", "scope": "Balanced lab subset"},
        {"dataset": "Iris", "path": "iris.csv", "records": len(iris), "experiments": "2", "source": iris_source,
            "scope": "Complete dataset"},
        {"dataset": "CIFAR-10", "path": "cifar10_lab.npz", "records": 10000, "experiments": "3, 4, Mini Project",
            "source": "TensorFlow Keras CIFAR-10", "scope": "Balanced lab subset"},
        {"dataset": "Dogs vs. Cats", "path": str(dogs_cats_dir.relative_to(dataset_dir)), "records": 3000,
            "experiments": "5", "source": "microsoft/cats_vs_dogs public mirror (Hugging Face)", "scope": "Filtered
            dataset (rebuilt 1000/1000 train, 500/500 validation)"},
        {"dataset": "Airline Passengers", "path": "monthly_airline_passengers.csv", "records": len(airline),
            "experiments": "6", "source": airline_url, "scope": "Complete dataset"},
        {"dataset": "MS COCO", "path": "ms_coco_subset", "records": 5, "experiments": "7", "source":
            "mm-eval/MS-COCO-Captions public parquet mirror", "scope": "Real-image demonstration subset"},
        {"dataset": "CelebA", "path": "celeba_subset", "records": celeba_count, "experiments": "8", "source":
            "flwrlabs/celeba public mirror", "scope": "Streamed training subset"},
        {"dataset": "Shakespeare Corpus", "path": "shakespeare.txt", "records":
            len(shakespeare_path.read_text(encoding="utf-8")), "experiments": "9", "source": "TensorFlow Tiny
            Shakespeare", "scope": "Complete corpus"},
        {"dataset": "LSUN Bedrooms", "path": "lsun_bedrooms_subset", "records": lsun_count, "experiments": "10",
            "source": "pcuenq/lsun-bedrooms public mirror", "scope": "Streamed training subset"},
    ]
)
manifest.to_csv(dataset_dir / "dataset_manifest.csv", index=False)

print(f"Deeplearning root: {root}")
print(manifest.to_string(index=False))
print("Dataset evidence:")
for path in sorted(dataset_dir.iterdir()):
    if path.is_dir():
        file_count = sum(1 for item in path.rglob("*") if item.is_file())
        print(f"{path.name}/: {file_count} files")
    else:
        print(f"{path.name}: {path.stat().st_size} bytes")

Deeplearning root: /content/Deeplearning
           dataset                              path  records        experiments
                                                
```

Procedure: NumPy forward propagation, backpropagation, training history, test accuracy, and sample predictions.

Run the cell and wait for the completion message. The model, numerical results, and plots are written to the corresponding folder under `/content/Deeplearning/outputs`.

Code

```
from pathlib import Path
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd


root = Path("/content/Deeplearning") if Path("/content").exists() else Path(__file__).resolve().parents[1]
dataset_dir = root / "datasets"
output_dir = root / "outputs" / "experiment_01"
output_dir.mkdir(parents=True, exist_ok=True)

with np.load(dataset_dir / "mnist_lab.npz") as dataset:
    X_train = dataset["x_train"].reshape(-1, 784).astype("float32") / 255.0
    y_train = dataset["y_train"].astype("int64")
    X_test = dataset["x_test"].reshape(-1, 784).astype("float32") / 255.0
    y_test = dataset["y_test"].astype("int64")

rng = np.random.default_rng(42)
input_size = 784
hidden_size = 64
output_size = 10
W1 = rng.normal(0, np.sqrt(1 / input_size), size=(input_size, hidden_size)).astype("float32")
b1 = np.zeros((1, hidden_size), dtype="float32")
W2 = rng.normal(0, np.sqrt(1 / hidden_size), size=(hidden_size, output_size)).astype("float32")
b2 = np.zeros((1, output_size), dtype="float32")


def sigmoid(values):
    values = np.clip(values, -30, 30)
    return 1.0 / (1.0 + np.exp(-values))


def softmax(values):
    shifted = values - values.max(axis=1, keepdims=True)
    exponentials = np.exp(shifted)
    return exponentials / exponentials.sum(axis=1, keepdims=True)


def forward(values):
    hidden = sigmoid(values @ W1 + b1)
    probabilities = softmax(hidden @ W2 + b2)
    return hidden, probabilities


one_hot = np.eye(output_size, dtype="float32")[y_train]
epochs = 12
batch_size = 128
learning_rate = 0.35
history_rows = []

for epoch in range(epochs):
    permutation = rng.permutation(len(X_train))
    epoch_loss = 0.0
    epoch_correct = 0
    processed = 0
    for start in range(0, len(X_train), batch_size):
        batch_indices = permutation[start : start + batch_size]
        batch_X = X_train[batch_indices]
        batch_y = one_hot[batch_indices]
        hidden, probabilities = forward(batch_X)
        loss = -np.mean(np.sum(batch_y * np.log(probabilities + 1e-9), axis=1))
        output_gradient = (probabilities - batch_y) / len(batch_X)
        W2_gradient = hidden.T @ output_gradient
        b2_gradient = output_gradient.sum(axis=0, keepdims=True)
        hidden_gradient = (output_gradient @ W2.T) * hidden * (1.0 - hidden)
        W1_gradient = batch_X.T @ hidden_gradient
        b1_gradient = hidden_gradient.sum(axis=0, keepdims=True)
        W2 -= learning_rate * W2_gradient
        b2 -= learning_rate * b2_gradient
        W1 -= learning_rate * W1_gradient
        b1 -= learning_rate * b1_gradient
        epoch_loss += float(loss) * len(batch_X)
        epoch_correct += int((probabilities.argmax(axis=1) == y_train[batch_indices]).sum())
        processed += len(batch_X)
    row = {
        "epoch": epoch + 1,
        "loss": epoch_loss / processed,
```

```
        "training_accuracy": epoch_correct / processed,
    }
    history_rows.append(row)
    print(row)

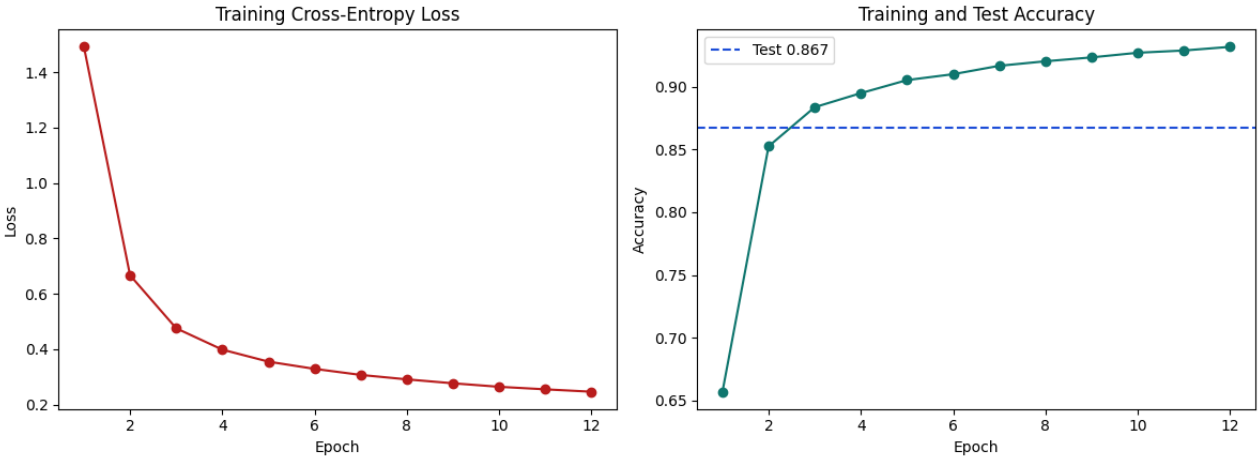
_, test_probabilities = forward(X_test)
test_predictions = test_probabilities.argmax(axis=1)
test_accuracy = float((test_predictions == y_test).mean())
history = pd.DataFrame(history_rows)
history["test_accuracy"] = test_accuracy
history.to_csv(output_dir / "training_history.csv", index=False)
np.savez_compressed(output_dir / "three_layer_network_weights.npz", W1=W1, b1=b1, W2=W2, b2=b2)

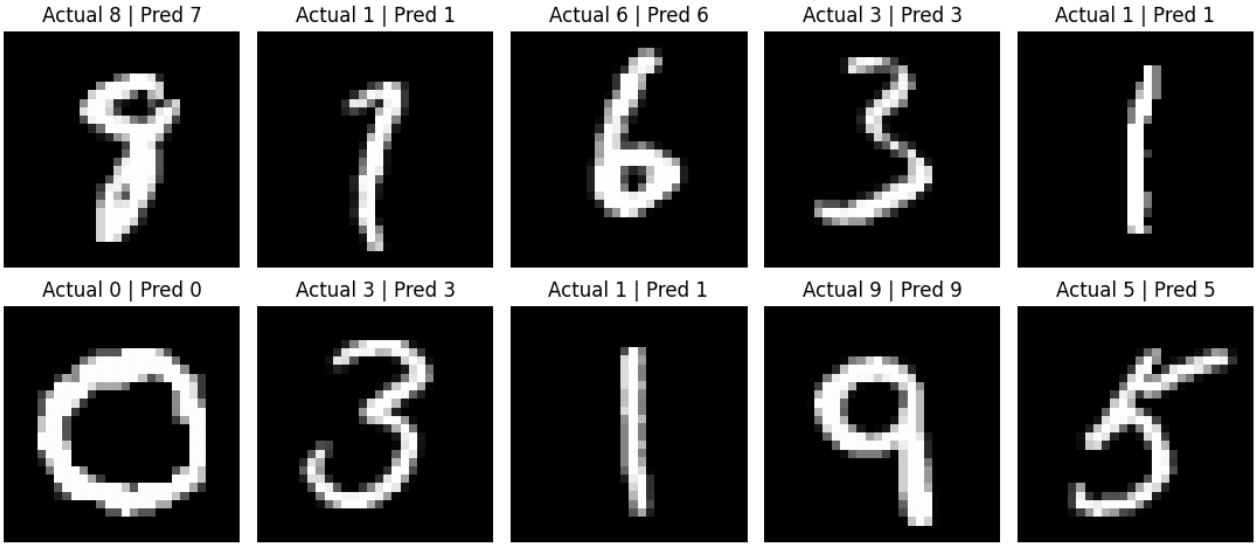
figure, axes = plt.subplots(1, 2, figsize=(12, 4.5))
axes[0].plot(history["epoch"], history["loss"], marker="o", color="#b91c1c")
axes[0].set_title("Training Cross-Entropy Loss")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("Loss")
axes[1].plot(history["epoch"], history["training_accuracy"], marker="o", color="#0f766e")
axes[1].axhline(test_accuracy, color="#1d4ed8", linestyle="--", label=f"Test {test_accuracy:.3f}")
axes[1].set_title("Training and Test Accuracy")
axes[1].set_xlabel("Epoch")
axes[1].set_ylabel("Accuracy")
axes[1].legend()
figure.tight_layout()
figure.savefig(output_dir / "training_performance.png", dpi=180, bbox_inches="tight")
plt.show()

sample_figure, sample_axes = plt.subplots(2, 5, figsize=(11, 5))
for index, axis in enumerate(sample_axes.ravel()):
    axis.imshow(X_test[index].reshape(28, 28), cmap="gray")
    axis.set_title(f"Actual {y_test[index]} | Pred {test_predictions[index]}")
    axis.axis("off")
sample_figure.tight_layout()
sample_figure.savefig(output_dir / "sample_predictions.png", dpi=180, bbox_inches="tight")
plt.show()

print("Experiment 1 completed")
print(f"MNIST training samples: {len(X_train)}")
print(f"MNIST testing samples: {len(X_test)}")
print(f"Test accuracy: {test_accuracy:.4f}")
print(f"Output folder: {output_dir}")

{'epoch': 1, 'loss': 1.4916087520599366, 'training_accuracy': 0.6569}
{'epoch': 2, 'loss': 0.6668563804626465, 'training_accuracy': 0.8526}
{'epoch': 3, 'loss': 0.4755387357234955, 'training_accuracy': 0.8838}
{'epoch': 4, 'loss': 0.39838216247558594, 'training_accuracy': 0.895}
{'epoch': 5, 'loss': 0.35502114095687864, 'training_accuracy': 0.9054}
{'epoch': 6, 'loss': 0.32897053065299986, 'training_accuracy': 0.9101}
{'epoch': 7, 'loss': 0.30712303256988527, 'training_accuracy': 0.9168}
{'epoch': 8, 'loss': 0.2910750477313995, 'training_accuracy': 0.9204}
{'epoch': 9, 'loss': 0.2768934510111809, 'training_accuracy': 0.9235}
{'epoch': 10, 'loss': 0.26427045230865476, 'training_accuracy': 0.9272}
{'epoch': 11, 'loss': 0.2550095099067688, 'training_accuracy': 0.929}
{'epoch': 12, 'loss': 0.24646899499893188, 'training_accuracy': 0.9319}
```






```
Experiment 1 completed
MNIST training samples: 10000
MNIST testing samples: 2000
Test accuracy: 0.8670
Output folder: /content/Deeplearning/outputs/experiment_01
```

Output check

Confirm that this experiment saved its results before moving on.

Code

```
from pathlib import Path
output_dir = Path('/content/Deeplearning/outputs/experiment_01')
files = sorted(path for path in output_dir.rglob('*') if path.is_file())
assert files, f'No outputs found in {output_dir}; run the experiment cell first.'
print(f'{len(files)} output files saved in {output_dir}')
for path in files:
    print(path.relative_to(output_dir))
```